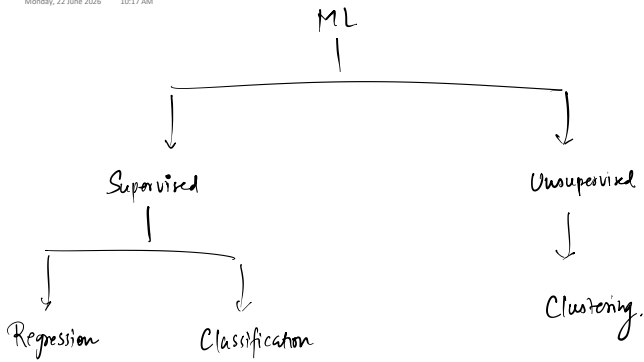




Supervised Learning Models :- Both Labels and Target are available

Unsupervised Learning Models :- Both Labels and Target are unavailable

Sr.No	Name	Father's Name	Ph. No.	email	Marks 1 st	Marks 2 nd	Marks 3 rd
					-	-	-
					-	-	-

(X)

All columns are called Features of datasets

The feature which we are going to predict is Target. (Y)

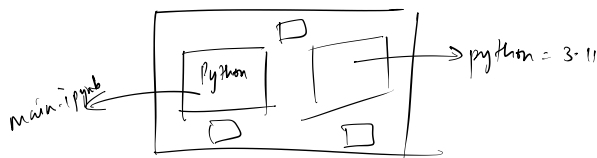
Column = features

rows = data points

predicted column = Target.

① If Target is continuous numerical value, regression model can be implemented.

② If Target is categorical data, we can implement any classification algorithm.



Debugging

↳ ① "Python" is not recognised command

Sol

↳ Search for "Environment Variables"

① Edit System Environment variables

Add a new entry to Path

Create a virtual env

python -m venv venv

Activate the virtual env.

for mac. source venv/bin/activate

for windows venv/Scripts/activate.ps1

or

cd venv

cd Scripts

./activate.ps1

Exploratory Data Analysis

- ↳ No. of rows
- No. of columns [features]
- Null values
- Duplicate values
- data type

Descriptive Analysis

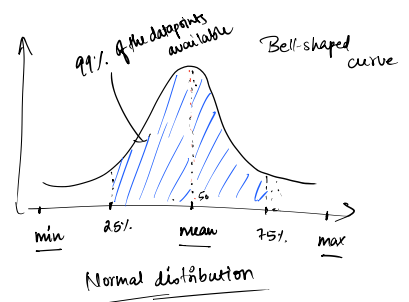
• info()

Statistical analysis

• describe()

↓
mean
count
std
min
max

{ 25%
50%
75% }



Any extreme value / unexpected value in the data can be considered as Outlier.

To build accurate ML model we need to remove this outlier.

Inter Quartile Range (IQR) = 75% - 25%.

$$IQR = Q_3 - Q_1$$

Acceptable value =

$$\text{Lower Bound} = Q_1 - 1.5 IQR$$

$$\text{Upper Bound} = Q_3 + 1.5 IQR$$

Others are considered as Outliers